Computer Representation of set :-

Bit string :- A bit is a digit which is either zero or one

Byte :- Byte is a string of 8 bits.

1	0	1	1	0	0	1	0

Make length 4 of string $= 2^4$ (2^n)

If $2^3 = 8$ then we get 8 bit string of length 3.

If $2^1 = 2$

0	→ string 1
1	→ string 2

If $2^2 = 4$

1 0	1 1	→ 4 bit string of length 2
0 1	0 0	

Computer representation of sets :-

There are several ways to represents sets using computers. One of them is to store the elements of the set in an unordered fashion. However, if this is done, the operation of computing the union, intersection or difference of two sets would be time consuming because each of these operations would require a large amount of searing for elements.

So, we will present a method for storing elements using an arbitrary ordering for elements of the universal set. This method makes easy for computing.

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Basic set operations Through Bit string Representation

Let the universal set be $U = \{1, 2, 3, 4, 5\}$

If $A = \{2, 4, 5\}$ and $A \subset U$

$\therefore$ Bit string of A is 01011

OR operation

$$\left.\begin{array}{l} 1 \text{ OR } 0 \\ 0 \text{ OR } 1 \\ 1 \text{ OR } 1 \end{array}\right\} 1$$

0 OR 0 $\rightarrow$ 0

AND operation

1 AND 1 $\rightarrow$ 1

$$\left.\begin{array}{l} 1 \text{ AND } 0 \\ 0 \text{ AND } 1 \\ 0 \text{ AND } 0 \end{array}\right\} \rightarrow 0$$

NOT operation

NOT 1 $\rightarrow$ 0

NOT 0 $\rightarrow$ 1

eg:- ① If $A = \{2, 4, 5\}$, $B = \{1, 3, 5\}$

Find ~~A∩B~~ bit string of $A \cap B$

Solution:-

~~A∩B =~~

Bit string of A = 01011

Bit string of B = 10101

$\therefore$ $A \cap B$ = 00001

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(e) If $A = \{2, 4, 5\}$, $B = \{1, 3, 5\}$ using bit string find $A \cup B$

Solution:- Bit string of $A = 01011$

Bit string of $B = 10101$

$\therefore A \cup B = 11111$

Complement of Bit string:

eg:- If $U = \{1, 2, 3, 4, 5\}$, $A = \{2, 4, 5\}$ using bit string find the complement of A.

Solution:- Bit string of A is 01011

$\therefore \overline{A} = \overline{01011} = 10100$

If $U = \{1, 2, 3, 4, \ldots, n\}$, how many subsets of U

We know each subset is represented by n bits and each bit is represented by 2 elements i.e. 0 and 1

$\therefore$ No. of subsets in $U = 2^n$

Cardinality of the power set of U.

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Ex. If $U = \{1, 2, 3, 4, 5, 6\}$, $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$. Find the bit string for the set A and B and use bit string to find union and intersection of A and B.

Solution:- Bit string of A is 111100

Bit string of B is 001111

$\therefore$ Bit string of $A \cup B$ is

$$\begin{array}{r} 111100 \\ 001111 \\ \hline 111111 \end{array}$$

Again, Bit string of $A \cap B$ is

$$\begin{array}{r} 111100 \\ 001111 \\ \hline 001100 \end{array}$$

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